

A Multi-Cohort Decision-Evaluation Benchmark for EHR World-Model Components and Offline Reinforcement Learning

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ABSTRACT

Electronic health record (EHR) world-model components are increasingly connected to offline reinforcement learning (RL), yet comparisons can be driven by incompatible tasks, actions, rewards, support rules, or evaluator assumptions rather than algorithmic differences. We introduce a development benchmark that standardizes these layers across six ICU task constructions and releases complete results for six transition methods, five uncertainty-capable methods, 34 policy/planner labels, and nine off-policy evaluation (OPE) estimators. Across the full transition inventory, no method is a global winner: the Transformer has the lowest one-step error on all six tasks, whereas final-horizon error is led by the Gaussian ensemble on five and persistence/LOCF on one; recursive CRPS is led by the ensemble, while recursive calibration error is led by GRU-D. In four adaptive exact-finite environments, the exact adaptive policy exceeds every fixed action, but a prespecified severity rule has the lowest regret among all 34 non-oracle labels; the learned-method leader is world-model-based on three tasks and model-free on one. Earlier monotone environments are retained only as evaluator smoke/sensitivity tests. OPE decisions use policy-set interval inclusion, rank, sign, false-improvement, effective-sample-size, support, and precision gates, but no complete retrospective EHR scoring contract is authorized. We therefore report no retrospective EHR policy value or winner. The contribution is an executable multi-cohort scaffold, an all-method performance inventory, and a failure decomposition from P/R/T component forecasting through policy-evaluation authorization.

CCS CONCEPTS

• **Applied computing** → **Health informatics**; • **Computing methodologies** → **Reinforcement learning**.

KEYWORDS

electronic health records, world models, offline reinforcement learning, benchmark, off-policy evaluation, uncertainty

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1 INTRODUCTION

EHR prediction benchmarks standardize cohorts, preprocessing, splits, and metrics [4, 14]. Offline RL over EHR trajectories adds a more fragile technical chain: recorded state must precede the action label, reward must follow it without mechanically encoding it,

policy probability must remain inside empirical support, learned dynamics must remain stable under recursion, and OPE must recover known values before retrospective scoring is authorized. Sepsis RL established a widely reused fluid-by-vasopressor action abstraction [6]; recent work connects latent EHR dynamics and imputation to policy learning [15]. These studies motivate a shared evaluator, but their different task and scoring contracts prevent direct algorithmic comparison.

Our primary contribution is the standardized comparison and failure decomposition across EHR P/R/T component learning, component model $\times$ planner combinations, model-based and model-free offline RL, known-value truth, retrospective support diagnostics, and cross-cohort evaluability. Transition accuracy is a validity layer rather than the endpoint. Temporal leakage control is a methods requirement rather than the headline contribution.

Because no genuinely untouched role remains, all EHR findings are labeled *development benchmark evidence*. The benchmark evaluates P/R/T component and model-free offline-RL pipelines across heterogeneous EHR tasks, compares policy behavior against exact truth in constructed adaptive environments, and audits support, collapse, planning stability, and model exploitation on retrospective EHR data. The retrospective layer evaluates computational behavior, not policy value.

Contributions. We provide: (1) a common, time-ordered task scaffold with train-only fitting and subject-disjoint development roles; (2) an AI-Clinician-aligned K25 sepsis scaffold plus five task-specific extensions; (3) complete all-method transition, uncertainty, policy, and OPE inventories rather than winner-only summaries; (4) a component-level bridge from prediction and uncertainty to known-value policy behavior and model exploitation; (5) exact OPE metrics and authorization decisions rather than estimator names alone; and (6) explicit non-execution when adaptive-environment, probability, support, or evaluator prerequisites fail.

Research questions. The benchmark asks: **RQ1**, how do P/R/T component families differ in one-step, recursive, reward, termination, uncertainty, and action-information fidelity? **RQ2**, do known-value environments require adaptive decisions before model-based and model-free policies are compared? **RQ3**, when do learned models exhibit exploitation gaps, and which OPE contracts recover known policy values and ranks? **RQ4**, which parts of this decision scaffold remain operational across heterogeneous cohorts without transferring sepsis-specific reward semantics?

2 RELATED DECISION BENCHMARKS

Table 1 separates task-specific retrospective EHR RL, EHR component/simulation models, and synthetic known-value benchmarks.

It is representative rather than systematic: the purpose is to expose which evidence layers are connected under one contract, not to claim exhaustive coverage.

Prior work occupies distinct evidence surfaces. Real-EHR studies usually use task-specific retrospective evaluators; MedDreamer applies one method scaffold to two task-specific contracts; general EHR models may stop at trajectory simulation; and synthetic benchmarks supply known returns without validating real-EHR task construction. We connect six real-EHR task contracts to common P/R/T diagnostics, matched policy inventories, known-value evaluator tests, and explicit retrospective authorization boundaries. Our respiratory and shock tasks remain published-inspired adapters rather than reproductions of DeepVent or POPCORN. Healthcare offline-RL model selection remains sensitive to reward, support, and estimator choice [9, 13].

3 BENCHMARK SCAFFOLD

3.1 State, action, target, and roles

Each transition uses a fixed four-hour clock. Let

$$x_t = [v_t, m_t, \delta_t, c, a_{t-1}], \quad y_t = v_{t+1}, \quad (1)$$

where v_t is the pre-action value vector, m_t its observation mask, δ_t feature-wise recency, c static/pre-anchor context, and a_{t-1} the previous action abstraction. Current action, post-action measurements, future masks, post-outcome fields, and whole-stay diagnosis proxies are prohibited from x_t . The logged action a_t summarizes recorded exposure or settings over $[t, t + 4h)$; y_t begins at $t + 4h$. Every evaluated bin remains inside one ICU stay.

Throughout, $K = |\mathcal{A}|$ denotes the number of discrete action classes, not the number of observed transitions or treatment dimensions.

The interval label is deliberately called a *recorded-exposure action*: it can include adaptation during the interval and is not asserted to be fully assignable at its start. It is valid for logged-action-conditioned forecasting and retrospective policy-behavior diagnostics. In the known-value environments, by contrast, a_t is an assignable intervention in the constructed transition kernel. This distinction prevents the real-EHR label from inheriting the stronger semantics of the simulator action.

Imputation, normalization, action cutpoints, support masks, feature selection, behavior denominators, hyperparameters, and checkpoints are fitted or selected on development train/validation roles only. The roles are subject disjoint, but historical use of the available evaluation roles precludes calling any role a new confirmatory holdout. No row is used to claim a confirmatory best method.

3.2 AI-Clinician-aligned K25 sepsis scaffold

Sepsis is aligned to the AI Clinician lineage only at the four-hour fluid-quintile $\times$ vasopressor-quintile decision scaffold ($K = 25$). It uses local 33- and 40-variable views with masks/time gaps, prior action, and locally reconstructed SOFA/lactate reward components. It is not an exact reproduction: numerical reward coefficients and parts of the source cohort implementation are not verifiable from the current artifact; the repaired anchor and stay-bounding rules

intentionally differ; the local SOFA proxy can mechanically overlap vasopressor action; and anchor-relative 90-day outcome ascertainment is unavailable. We therefore retain an aligned scaffold and treatment-isolated reward sensitivity, not a canonical reproduction.

The matched feature/reward/denominator comparison did not support choosing a preferred setting because terminal outcome availability and matched semi-synthetic reliability were incomplete. The K25 scaffold remains a compatibility surface despite weak factual action information.

3.3 Task-specific extensions

Table 2 summarizes extension roles and aggregate scale. An episode is one materialized ICU stay in the subject-disjoint development train or validation role; episodes and transitions should not be summed across cohorts because one person or stay may satisfy more than one task definition. Respiratory uses observed PEEP quintile $\times$ observed FiO₂ quintile ($K = 25$); a row is excluded when either setting is unavailable rather than mapped to no action. Its primary dense benchmark reward is

$$r_t^{\text{resp}} = \mathbb{I}[94 \leq \text{SpO}_{2,t+1} \leq 98] - .5\mathbb{I}[\text{SpO}_2 \text{ outside}] + \mathbb{I}[70 \leq \text{MBP}_{t+1} \leq 80] - .5\mathbb{I}[\text{MBP outside}], \quad (2)$$

with an SpO₂-only sensitivity. Shock uses fluid quintile $\times$ vasopressor quintile and two separated reward components,

$$r_t^{\text{MAP}} = \text{clip}((\text{MBP}_{t+1} - 65)/25, -1, 1), \quad (3)$$

$$r_t^{\text{lac}} = -\tanh(\text{lac}_{t+1} - \text{lac}_t).$$

All are frozen post-action benchmark objectives rather than transferred sepsis semantics. Sepsis world-model reward diagnostics use the lactate component; its reference terminal outcome is unavailable. AF/flutter remains world-model-only because strict action-reward ordering is unresolved. AKI and heart failure retain their original P/T transition surfaces and add separate decision-task extensions: severe AKI uses daily wait versus initiate RRT ($K = 2$, absorbing after initiation, at most three decisions), while heart failure uses a six-hour loop-diuretic-quartile by vaso/inotrope indicator action ($K = 8$, at most 12 decisions). Their post-action reward proxies define task-matched known-value mechanisms; they do not create retrospective EHR policy values.

4 METHODS COMPARED

4.1 P/R/T component models and metrics

The evaluated component is

$$\hat{M} = (\hat{P}_\theta(v_{t+1} | x_t, a_t), \hat{R}, \hat{T}), \quad (4)$$

not a complete generator of the future observation process. The core includes recurrent/GRU-D, causal Transformer, categorical recurrent state-space, and Gaussian recurrent ensemble models; persistence and histogram gradient boosting are sanity controls. Neural runs use seeds 3408/3411/3414, hidden dimension 48, latent dimension 16 where applicable, batch size 512, learning rate 0.002, and weight decay 10^{-4} . Checkpoints minimize 0.7 RMSE+0.3 MAE on development validation after train-only fitting, with 8 minimum and 40 maximum epochs, patience 6, and one uniform 80-epoch extension when the frozen convergence trigger fires.

Table 1: Representative landscape of EHR decision-model evaluation. “Known return” means value supplied by the environment rather than estimated from retrospective trajectories. Multisite evaluation is separated from multiple decision-task contracts.

Study	Data and decision task	Model / policy layer	Evaluation evidence	Cross-task scope
AI Clinician [6]	MIMIC-III + eRI; sepsis fluid-vasopressor decisions	Finite MDP + policy iteration	Retrospective OPE and action-outcome concordance; no known return	One task; two datasets
POPCORN [2]	MIMIC-III acute hypotension; hourly K20 fluid-vasopressor actions	Prediction-constrained POMDP + PBVI	Forecasting/likelihood + CWPDIS, ESS, and support; no known return	One task
DeepVent [7]	MIMIC-III; four-hour ventilation settings	Model-free CQL	FQE and action-distribution diagnostics; no known return	One task
MedDreamer [15]	MIMIC sepsis + eICU ventilation	Latent RSSM + imagination-based policy learning	Learned-reward/relative-mortality metrics + CWPDIS/ESS; no known return	Two tasks; task-specific actions/rewards
EHR-MPC [12] (preprint)	Eight-hospital ICU data; sepsis policy evaluation	Generative EHR twin + MPC	Importance weighting + learned-twin simulation; no known return	One policy task; multiple sites
EHRWorld [11] (preprint)	MIMIC-IV longitudinal trajectories	Intervention-aware long-horizon model; no policy learner	Forecasting and simulation fidelity	No policy benchmark
EpiCare / DTR-Bench [3, 10]	Synthetic clinical decision tasks	Multiple offline and online RL methods	Simulator true return + OPE	Multiple synthetic tasks; no real-EHR task construction
Ours	MIMIC-IV; six real-EHR task contracts	Four P/R/T component families + matched policy layers	Adaptive exact-finite truth + monotone sensitivity + retrospective authorization boundaries; no EHR winner	Shared cross-task contract

Table 2: Development task scale and evaluation layers. An episode is one materialized ICU stay; transitions are fully contained four-hour state-action-target rows. “Supported/K” gives train-supported EHR action classes over the discrete action space. AKI and heart failure additionally use the named decision extensions. No row has retrospective EHR policy value.

Cohort	Episodes	Transitions	Action contract	Benchmark role	Reward / value status
Sepsis	3,440	34,245	22/25 fluid-vasopressor	K25 sepsis scaffold	P/R/T plus support/collapse diagnostics
Respiratory support	21,679	125,246	12/25 PEEP-FiO ₂	Policy extension	P/R/T plus known-value sensitivity
Shock	49,495	390,341	25/25 fluid-vasopressor	Policy extension	P/R/T plus known-value sensitivity
AF/flutter	4,276	33,836	16/25 rate-rhythm	Component-only	P/T; action-reward ordering unresolved
AKI	21,083	181,311	P/T: K4 compact; policy: K2 wait/RRT	Decision extension	P/T plus known-value sensitivity
Heart failure	3,986	32,198	P/T: K2 compact; policy: K8 diuretic × vaso/inotrope	Decision extension	P/T plus known-value sensitivity

The Gaussian ensemble combines the three recurrent members and is one derived row per task.

One-step RMSE/MAE uses only observed target cells after train-fitted normalization. Recursive rollout starts from the first validation state, feeds each predicted value back, marks generated future values as unobserved, increments recency, and follows the logged action sequence for eleven four-hour steps. At each horizon, error is scored against cells observed at that step in the recorded target mask. The scalar recursive error is the normalized observed-cell RMSE at the final executed horizon (step 11, 44 hours), averaged across available executed-seed rows for a method-cohort pair; all 924 horizon-resolved rows are released. Thus recursion tests value-state compounding under a fixed missingness convention; it does not claim to generate future measurement decisions. Reward heads are deterministic mappings from predicted state when available. Termination is evaluated natively, after train-only recalibration, and against a train empirical-hazard baseline.

Uncertainty reporting includes Gaussian NLL/CRPS, coverage at 50, 80, 90, and 95%, interval width and score, and risk-coverage. Factual action information uses one fixed primary estimand throughout: shuffled-action minus observed-action RMSE on reward-relevant targets within matched strata, averaged over the four core model families and executed seeds. Each model-seed contrast also carries a 500-replicate materialized-episode bootstrap interval. It is a predictive-information diagnostic.

4.2 Planners and policies

The planner matrix uses exact supported-action search at horizon one and categorical cross-entropy method (CEM) planning at nominal horizons four and eight. Each CEM iteration evaluates 64 sequences, retains eight elites, updates categorical probabilities with smoothing 0.2, applies the frozen support mask at every step, and executes only the first action before replanning. Three iterations and matched model-query budgets are used. The effective horizon is truncated by the remaining task horizon; therefore AKI’s nominal H4 and H8 variants are both H3-effective and are retained only as budget-contract diagnostics. The uncertainty sensitivity subtracts 0.1 times predictive scale.

The model-free core includes behavior cloning, independent implementations of discrete BCQ [1] and CQL [8], a Soft-SPIBB conceptual adapter, a Decision Transformer contract adapter, and supported random, minimum, maximum, and severity-rule controls. Neural runs use three fixed seeds, one train/validation trial, at most 20,000 training windows, 12 epochs, hidden dimension 64, batch size 512, and discount 0.99. Fidelity labels distinguish independent reimplementations from conceptual adapters. Deterministic controls are prespecified ϵ -soft distributions over the supported set with $\epsilon = 0.001$ for retrospective probability diagnostics; they are not scored by retrospective OPE.

4.3 Adaptive known-value and OPE evaluation

The primary policy surface uses four exact finite, known-construction environments matched to the respiratory, shock, AKI-RRT, and heart-failure action counts, horizons, supported sets, behavior concentration, missingness, and termination prevalence. Their transition/reward kernels contain low-state action harm, high-state under-treatment, intermediate optima, action cost, delayed benefit, response heterogeneity, switch cost, and dense-terminal trade-offs. The respiratory, shock, and AKI environments have 49 states and heart failure has 61. Dynamic programming supplies the exact adaptive policy and value. A preflight gate requires this adaptive policy to exceed the best fixed action by at least 0.02, more than one nontrivial optimal action, zero negative exact regret, valid support, and null-response equality. All four tasks pass before learned policies are inspected.

The earlier respiratory/shock reference family uses 13 states, 25 actions, 11 decisions, and a monotone dose-response. Task-matched AKI and heart-failure variants similarly retain transparent monotone controls. Because these mechanisms structurally favor fixed extreme actions, their results are now evaluator smoke/sensitivity tests rather than the primary policy comparison. They remain released to preserve provenance and to test support masks, common random numbers, planner execution, and OPE implementations.

For both adaptive and sensitivity families, all action responses, rewards, and termination mechanisms are known by construction. World-model architectures and policy methods are retrained within each environment; EHR-trained weights are not evaluated in these simulators. Adaptive experiments use five fixed seeds and compare five controls, five model-free methods, 24 world-model-planner combinations, and the exact oracle. Fixed-action controls are distinct from the state-dependent severity rule.

All policies share initial states and exogenous noise streams, enabling paired differences. The reported behavior-relative quantity is the Monte Carlo mean of $G^\pi - G^{\pi_b}$ under common environment randomness; stochastic action streams are independently seeded and recorded. Consequently, an empirical-behavior policy can have a small finite-sample paired difference even though its exact value equals the behavior reference. Each result averages three evaluation seeds and reports the paired Monte Carlo standard error. The released AKI/HF matrix also retains exact dynamic-programming value differences in separate columns; these are not mixed with the Monte Carlo standard errors used in the main table. Exact finite regret uses dynamic programming; semi-synthetic evaluation reports true policy returns without calling a high-budget planner an oracle.

The OPE matrix tests trajectory IS and WIS; per-decision IS, WPDIS, and CWPDIS; DR and WDR; FQE; and support-restricted variants. A contract fixes estimator, behavior denominator, clipping, support rule, horizon, and regime. The *policy-set interval inclusion rate* is the fraction of policy-specific bootstrap intervals containing each policy's known value within one fixed logged-data realization. It is not repeated-dataset frequentist coverage. Approval requires this diagnostic ≥ 0.90 ; Spearman rank and pairwise-order recovery ≥ 0.80 ; behavior-relative sign recovery ≥ 0.90 ; false-improvement rate ≤ 0.05 ; median ESS

$\geq \max(100, 0.05N)$; zero unsupported target mass; maximum paired true-return SE ≤ 0.03 ; and finite required values. No threshold is changed after inspection.

5 RESULTS

5.1 RQ1: P/R/T component fidelity

The full transition inventory comprises persistence/LOCF and HGB controls plus four world-model families. The Transformer has the lowest one-step RMSE on all six cohorts. At the final recursive horizon, the Gaussian ensemble leads five cohorts, while persistence/LOCF leads heart failure; when restricted to the four learned world-model families, the ensemble leads all six. Across the 24 core world-model-cohort bridge rows, normalized one-step error ranges from 0.669 to 0.773 and final-horizon error from 0.901 to 3.268. Thus the one-step selection reverses on every task. These are descriptive point leaders: neural methods have three seeds, the derived ensemble has one task-level row, and no subject-cluster leader interval is available.

Table 3 consolidates every executed comparison layer. It shows a stronger conclusion than a single aggregate leaderboard: some methods dominate one frozen metric, but the identity of the leader changes across point prediction, probabilistic recursion, adaptive policy comparison, and evaluator authorization. The complete layer-specific inventory and every granular result row are released.

Table 4 reports every transition method under the same task-balanced aggregation rather than only winners. Persistence/LOCF has no probabilistic output, and HGB has no matched action-information row; these missing cells are structural rather than suppressed results. The complete 36 cohort-method rows, including poor results, are released separately so users can reproduce a chosen cohort rather than reverse-engineer a headline leader.

Under the fixed action-information estimand, cohort means range from 0.0001 to 0.0023. Episode-cluster intervals exclude zero in all 10 model-seed rows for respiratory, shock, and AKI; 2/10 for AF/flutter; and 0/10 for sepsis and heart failure. Statistical separation can coexist with negligible magnitude and does not repair a missing repeated-action contract. The empirical-hazard termination baseline has lower Brier score than native and recalibrated learned heads in all 18 comparable K25-sepsis rows, so learned termination is a sensitivity rather than an unqualified planning component.

Uncertainty is also cohort- and metric-dependent. Across HGB and the four probabilistic world models, the one-step CRPS leader is the Transformer in five cohorts and the ensemble in sepsis. Under recursive rollout, the ensemble has the lowest CRPS in all six, whereas GRU-D has the lowest MACE in all six. The recursive 90% coverage range is especially broad in sepsis (0.623–0.941) and heart failure (0.657–0.934). Hence neither one-step point error nor one-step uncertainty identifies recursive sharpness and calibration jointly.

5.2 RQ2: Adaptive known-value policy performance

The adaptive environments pass the construction gate: the exact adaptive policy exceeds the best fixed action by 0.188–0.439

Table 3: End-to-end baseline and method atlas. CRPS, mean absolute coverage error (MACE), and exact regret are minimized. Point leaders are descriptive within a frozen layer and do not define a global cross-layer winner.

Layer	Metric	Evaluated inventory	Scope	Point-leader pattern
Transition	One-step RMSE	LOCF, HGB, GRU-D, Transformer, RSSM, ensemble (6)	6 EHR cohorts	Transformer 6/6
Transition	Final-horizon RMSE	Same six methods	6 cohorts; step 11 (44 h)	Ensemble 5/6; LOCF 1/6
Uncertainty	One-step CRPS / MACE	HGB plus four probabilistic WMs (5); LOCF unavailable	6 EHR cohorts	CRPS: Transformer 5/6, ensemble 1/6; MACE: Transformer 4/6, GRU-D 2/6
Uncertainty	Recursive CRPS / MACE	Same five probabilistic methods	6 EHR cohorts	CRPS: ensemble 6/6; MACE: GRU-D 6/6
Policy	Exact regret	5 controls + 5 model-free + 24 WM-planner labels (34)	4 adaptive exact-finite tasks; 680 non-oracle seed rows	Severity rule 4/4
OPE	Frozen authorization	IS, WIS, PDIS, WPDIS, CWPDIS, DR, WDR, FQE, support-restricted WPDIS (9)	Exact and aggregate-calibrated evaluator contracts	Tuple-specific synthetic approvals; none retrospective

Table 4: Complete task-balanced transition/component performance. Each value first averages executed seeds within a cohort and method, then takes an unweighted mean over all six cohorts. RMSE, CRPS, and MACE are minimized; Cov90 is empirical recursive 90% interval coverage. Δ_a is matched-shuffle minus observed-action RMSE on reward-relevant targets and is predictive, not causal. Full cohort-specific rows are released.

Group	Method	1-step RMSE	Recursive RMSE	1-step CRPS	Recursive CRPS	Rec. Cov90	Rec. MACE	Δ_a
Control	Persistence/LOCF	0.777	1.095	—	—	—	—	—
Control	HGB residual	0.695	1.018	0.354	0.479	0.827	0.060	—
P/R/T	GRU-D	0.688	1.107	0.341	0.485	0.904	0.023	0.0011
P/R/T	Transformer	0.683	2.579	0.337	0.839	0.710	0.164	0.0010
P/R/T	Categorical RSSM	0.749	1.161	0.371	0.527	0.795	0.088	0.0019
P/R/T	Gaussian ensemble	0.684	0.969	0.338	0.458	0.937	0.056	0.0008

across the four tasks. This removes the monotone-design failure in which an extreme action was optimal by construction. Each task then evaluates the same 34 non-oracle labels—five controls, five model-free methods, and 24 world-model-planner combinations—over five seeds with exact dynamic-programming values and no support-mask bypass.

Figure 1 reports every label. Define behavior-normalized exact regret as

$$\tilde{\mathcal{R}}(\pi) = \frac{V^* - V^\pi}{V^* - V^{\pi_b}}, \quad (5)$$

so the exact oracle is zero and behavior is one within each task. A prespecified state-dependent severity rule has the lowest regret among the 34 labels on all four tasks: 0.003, 0.124, 0.175, and 0.241 times behavior regret for AKI, heart failure, respiratory, and shock, respectively. No learned method beats that rule. Among learned methods, a world-model planner leads respiratory, AKI, and heart failure, whereas discrete BCQ leads shock; the selected world-model family, planner horizon, and uncertainty penalty vary by task. Thus the new experiment itself is valid, but its learned baselines fail to recover the simple adaptive control.

The historical monotone matrices are no longer used for method ranking. Their complete 2,448 method-seed-scenario rows and all 34-label task summaries remain available as smoke/sensitivity evidence, including poor methods, for evaluator regression testing.

5.3 RQ3: Exploitation and OPE disposition

The AI-Clinician-aligned K25 sepsis scaffold passes 98 of 1,800 horizon-specific pre-estimator gates. Material seed instability occurs in 26 of 50 rows, unsupported probability mass in 9 of 150, and marginal single-action collapse in 18 of 150. These failures explain why probability-complete training output is not sufficient for policy-value scoring.

World-model planners also exhibit substantial simulator-to-truth disagreement in the adaptive environments. Across 120 planner-seed rows per task, mean absolute learned-model-versus-exact-return gaps range from 0.377 to 2.693. These values are on task-specific constructed reward scales and are not compared across tasks; they show why learned-model value cannot substitute for exact known-value evaluation.

The shared reference evaluator approves 32 of 3,456 exact-finite tuples. All 32 are masked-support, moderate-regime, horizon-one contracts and span CWPDIS, DR, IS, PDIS, WDR, WIS, WPDIS, and support-restricted WPDIS. Table 5 shows why family-name transport is invalid: among 3,456 aggregate-calibrated semi-synthetic tuples, none passes policy-set interval inclusion, rank, or sign recovery, even though 3,264 have finite outputs and 1,728 pass support and precision individually.

Task matching changes the result rather than uniformly rescuing OPE. For the separate AKI evaluator, 236 of 1,728 semi-synthetic tuples pass all nine gates across weak, moderate, and delayed regimes and horizons one through three. For heart failure, 0 of 2,880 pass. The AKI result validates only those exact estimator-denominator-clip-support-horizon-regime tuples in the constructed environment. Retrospective AKI scoring is still not executed because this run has neither a probability-complete EHR-trained target policy nor its policy-specific post-training overlap/ESS receipt. Thus no estimator is transported by family name, no retrospective EHR OPE value is reported, and all 16,128 reference and task-matched tuple metric/disposition rows are released.

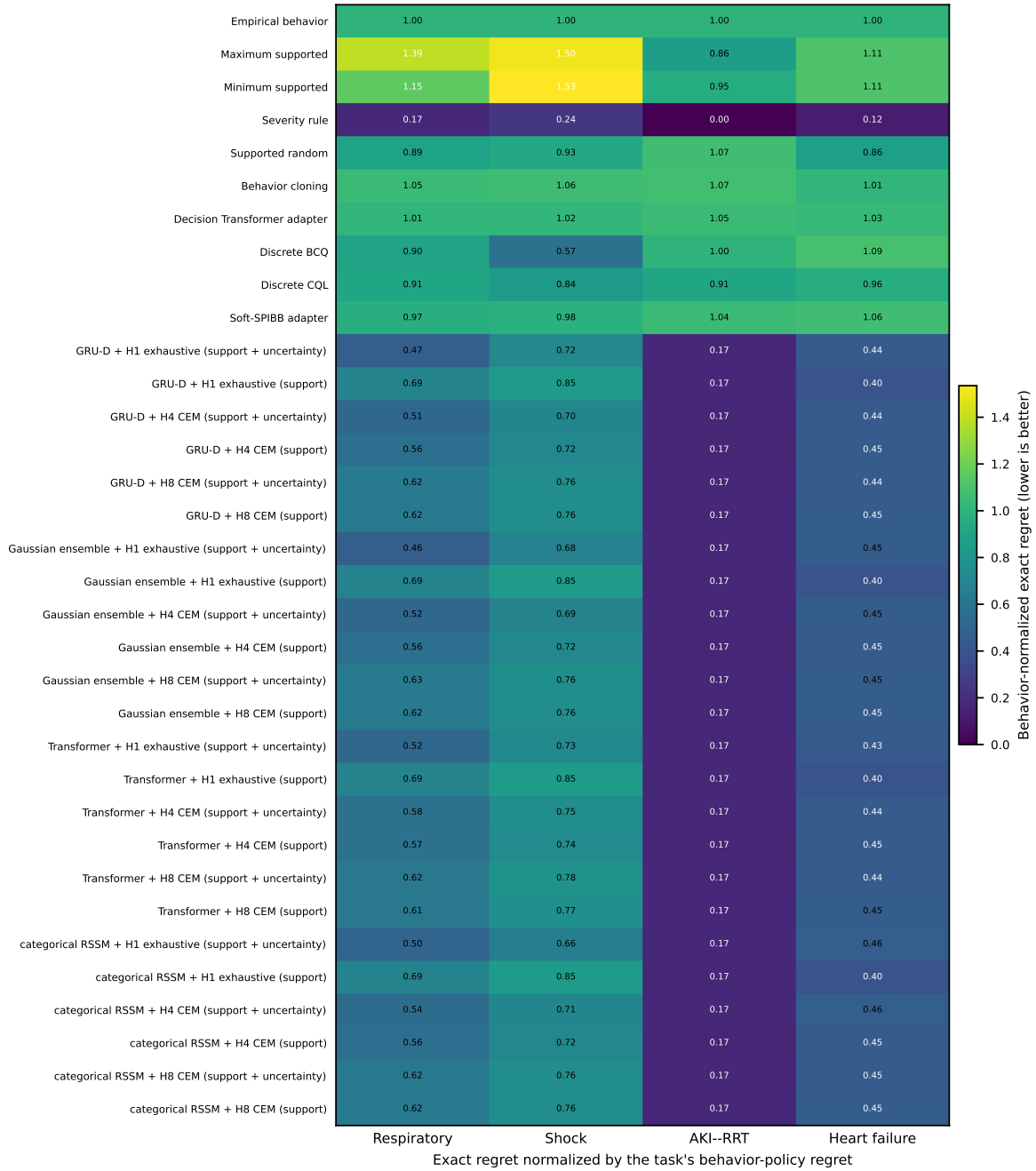


Figure 1: All 34 non-oracle policies in four adaptive exact-finite environments. Cells are five-seed mean exact regret normalized by behavior-policy regret (lower is better; oracle = 0, behavior = 1). Raw exact returns, regrets, seed variation, and unsupported mass are released for every cell. The severity rule leads each task; no learned method is a universal winner.

5.4 RQ4: Cross-cohort evaluability

The common scaffold produces time-ordered transition tasks in all six cohorts, but higher evaluation layers differ. Respiratory and shock retain rich repeated actions, post-action reward proxies, P/R/T components, and known-value policy matrices.

AF/flutter has a supported $K = 25$ transition action but fails strict action-reward ordering and remains world-model-only. AKI and heart failure require separate decision contracts rather than promotion of their compact transition actions: daily absorbing

Table 5: OPE gate pass counts; each tier has 3,456 estimator-denominator-clip-support-horizon-regime tuples. Policy-set inclusion is not nominal repeated-dataset coverage. Thresholds are frozen in Methods.

Gate	Exact finite	Aggregate-calibrated
Policy-set inclusion	108	0
Rank	144	0
Pairwise order	800	648
Improvement sign	137	0
False improvement	1,204	215
ESS	279	270
Support	1,728	1,728
Paired precision	1,728	1,728
Finite values	3,264	3,264
All nine	32	0

RRT initiation and six-hour loop-diuretic by vaso/inotrope, respectively. Both support known-value model-based/model-free comparison; only AKI yields task-matched semi-synthetic OPE candidates, and neither reaches retrospective EHR value scoring. The benchmark therefore reports a six-task world-model surface, four known-value policy surfaces, one world-model-only task, and zero retrospective policy-value surfaces.

6 PREDICTION-TO-POLICY FAILURE DECOMPOSITION

Prediction quality is neither ignored nor treated as policy performance. The machine-readable bridge links one-step and recursive errors, reward error, termination calibration, factual action information, support error, model-exploitation gap, known-value return, and retrospective scoring status for every core world model. Four patterns are important.

First, one-step rankings do not persist under recursive rollout: the Transformer-to-ensemble reversal occurs in five cohorts and Transformer-to-LOCF in heart failure; among the four learned P/R/T component families, the Transformer-to-ensemble reversal occurs in all six. Second, cohort-level uncertainty shows that recursive CRPS and recursive calibration select different methods. Third, adaptive learned-model-versus-exact-return gaps remain substantial, so simulated value cannot substitute for known truth. Finally, the adaptive oracle beats every fixed action, yet every learned method loses to the severity rule and the best learned family changes by task. The earlier extreme-action winners are therefore correctly interpreted as monotone-environment smoke results rather than policy findings. Exploratory prediction-policy correlations share environments and learned models and are not treated as independent evidence.

The resulting benchmark logic is compositional: a row may remain valid for P/R/T forecasting while failing planning discrimination, and a policy may remain valid for support/collapse analysis while failing value authorization. This prevents an upper-layer failure from erasing lower-layer evidence or a lower-layer success from silently authorizing a higher-layer claim.

7 REPRODUCIBILITY AND ACCESS

The reader package includes a benchmark card; task, feature, action, reward, planner, model, and estimator cards; sanitized policy-level and tuple-level result tables; a number-to-source audit; immutable source hashes; environment manifests; build commands; and credentialed reconstruction instructions. It exposes all 36 cohort-transition-method rows, task-balanced results for all six transition methods, all 680 adaptive non-oracle method-seed rows, all 2,448 monotone sensitivity rows, all OPE tuple metrics/dispositions, horizon-resolved transition results, cohort-specific uncertainty summaries, and gate thresholds rather than only headline aggregates. Code, contracts, synthetic fixtures, aggregate evidence, and clean-room commands are provided in the anonymous artifact at <https://anonymous.4open.science/r/ehrdyn-icu-65FB>. The source data are MIMIC-IV v3.1 [5]; access remains governed by PhysioNet credentialing and its data-use agreement. Raw EHR rows, identifiers, exact timestamps, trajectories, role-membership files, row-level predictions, tensors, and checkpoints are not redistributed.

Authorized users rebuild task arrays locally, fit all data-dependent transforms on development training roles, run development validation, execute exact and aggregate-calibrated known-value evaluation, and regenerate the aggregate synthesis and PDF. Unrestricted users can validate the packaged contracts and aggregate hashes and run the known-value tests without MIMIC-IV. Independent credentialed reconstruction remains pending: it is a reproducibility check in which another MIMIC-credentialed user regenerates aggregate results from a clean clone, not external-data validation, clinical validation, or a generalization experiment.

8 LIMITATIONS, DATA GOVERNANCE, AND SCOPE

The EHR findings are development benchmark evidence: no genuinely untouched confirmatory role is available. Component-model leader comparisons lack subject-cluster intervals, and the derived Gaussian ensemble contributes one row per task. Recursive evaluation predicts values under a fixed missingness convention rather than modeling a generative observation process; accordingly, we claim P/R/T components rather than a complete EHR world model with an observation head. The four-hour action is a recorded interval exposure, not guaranteed to be assignable at the bin start. Sepsis is an AI-Clinician-aligned K25 scaffold rather than an exact reproduction; its reference terminal outcome is unavailable. AKI and heart-failure policy comparisons use separately constructed known-value tasks rather than the original EHR transition actions, and AF remains component-only.

The adaptive policy mechanisms are exact and heterogeneous but constructed; their results do not establish value on retrospective EHR trajectories. The earlier monotone mechanisms remain implementation sensitivities only. AKI has task-matched aggregate-calibrated OPE approvals, but no task supplies the trained target-policy and post-training gates needed for retrospective scoring; the benchmark therefore does not rank policies on EHR data. MIMIC-IV is restricted, so the artifact shares code,

contracts, synthetic fixtures, and aggregate results rather than row-level data. These are evidence boundaries of the technical benchmark.

9 CONCLUSION

We present a common decision-evaluation scaffold that connects EHR P/R/T component learning, recursive uncertainty, planning, model-free RL, adaptive known-value truth, and OPE authorization. Its main empirical finding is not a universal winner: one-step, recursive, probabilistic, adaptive-policy, and estimator layers produce different rankings and failure modes. By publishing exact contracts, every method row, fixed controls, and non-execution rules, the benchmark makes those differences reusable while preserving valid lower-layer results. Retrospective EHR policy ranking remains unavailable until an exact evaluator contract passes the frozen bridge.

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